

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
APPLIED BUSINESS ANALYTICS PRACTICAL
Practical – IA

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Date of Assignment:24-07-26	Date/Time of Submission:24-07-26

Q1.Collect data from a real life business scenario and perform exploratory

Data analysis to gain insights into the dataset

Code - Output:-

Import data set and load csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Retail_sales.csv")

df.head()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
0	Spearsland	52372247	2022-01-01	9	2741.69	20	81	Tanzania	Furniture	Saturday	False
1	Spearsland	52372247	2022-01-02	7	2665.53	0	0	Mauritania	Furniture	Sunday	False
2	Spearsland	52372247	2022-01-03	1	380.79	0	0	Saint Pierre and Miquelon	Furniture	Monday	False
3	Spearsland	52372247	2022-01-04	4	1523.16	0	0	Australia	Furniture	Tuesday	False
4	Spearsland	52372247	2022-01-05	2	761.58	0	0	Swaziland	Furniture	Wednesday	False

Display last row

```
df.tail()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
29995	Spearsland	50239115	2022-01-25	5	2501.15	0	100	Malawi	Clothing	Tuesday	False
29996	Spearsland	50239115	2022-01-26	3	1500.69	0	0	Sudan	Clothing	Wednesday	False
29997	Spearsland	50239115	2022-01-27	6	3001.38	0	0	South Georgia and the South Sandwich Islands	Clothing	Thursday	False
29998	Spearsland	50239115	2022-01-28	5	2501.15	0	0	Haiti	Clothing	Friday	False
29999	Spearsland	50239115	2022-01-29	3	1425.66	5	190	Mozambique	Clothing	Saturday	False

Display shape database

```
df.shape
```

✓ 0.0s

(30000, 11)

Display column name

```
df.columns
```

✓ 0.0s

Index(['Store ID', 'Product ID', 'Date', 'Units Sold', 'Sales Revenue (USD)',
 'Discount Percentage', 'Marketing Spend (USD)', 'Store Location',
 'Product Category', 'Day of the Week', 'Holiday Effect'],
 dtype='str')

Display dataset Information

```
df.info()
```

[10] ✓ 0.0s

... <class 'pandas.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 11 columns):
Column Non-Null Count Dtype
--- ---
0 Store ID 30000 non-null str
1 Product ID 30000 non-null int64
2 Date 30000 non-null str
3 Units Sold 30000 non-null int64
4 Sales Revenue (USD) 30000 non-null float64
5 Discount Percentage 30000 non-null int64
6 Marketing Spend (USD) 30000 non-null int64
7 Store Location 30000 non-null str
8 Product Category 30000 non-null str
9 Day of the Week 30000 non-null str
10 Holiday Effect 30000 non-null bool
dtypes: bool(1), float64(1), int64(4), str(5)
memory usage: 2.3 MB

Display datatype

```
df.dtypes
✓ 0.0s
Store ID          str
Product ID       int64
Date             str
Units Sold       int64
Sales Revenue (USD) float64
Discount Percentage int64
Marketing Spend (USD) int64
Store Location    str
Product Category  str
Day of the Week   str
Holiday Effect    bool
dtype: object
```

Check missing Values

```
df.isnull().sum()
✓ 0.0s
Store ID          0
Product ID        0
Date              0
Units Sold        0
Sales Revenue (USD) 0
Discount Percentage 0
Marketing Spend (USD) 0
Store Location     0
Product Category   0
Day of the Week    0
Holiday Effect     0
dtype: int64
```

Check duplicate Record

```
df.duplicated().sum()
np.int64(0)
```

Descriptive statics

```
df["Product Category"].unique()
✓ 0.0s

<StringArray>
['Furniture', 'Electronics', 'Groceries', 'Clothing']
Length: 4, dtype: str
```

Check unique value

```
df.nunique()

Store ID          1
Product ID       42
Date            731
Units Sold       32
Sales Revenue (USD) 2545
Discount Percentage 5
Marketing Spend (USD) 200
Store Location   243
Product Category 4
Day of the Week  7
Holiday Effect   2
dtype: int64
```

Display particular columns

```
df["Product Category"].unique()

array(['Furniture', 'Electronics', 'Groceries', 'Clothing'], dtype=object)
```

Product category

```
df["Product Category"].value_counts()

Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

Store location

```
df["Product Category"].value_counts()
```

```
Store Location
Korea          237
Congo          234
Turks and Caicos Islands  156
Guam           153
Tanzania       151
...
Trinidad and Tobago    98
Haiti                96
Dominica             95
Ireland              92
Andorra              80
Name: count, Length: 243, dtype: int64
```

Holiday effect

```
df["Holiday Effect"].value_counts()
```

```
Holiday Effect
False    29836
True      164
Name: count, dtype: int64
```

Sales revenue

```
df["Sales Revenue (USD)"].mean()
df["Sales Revenue (USD)"].max()
df["Sales Revenue (USD)"].min()
```

✓ 0.0s

```
np.float64(0.0)
```

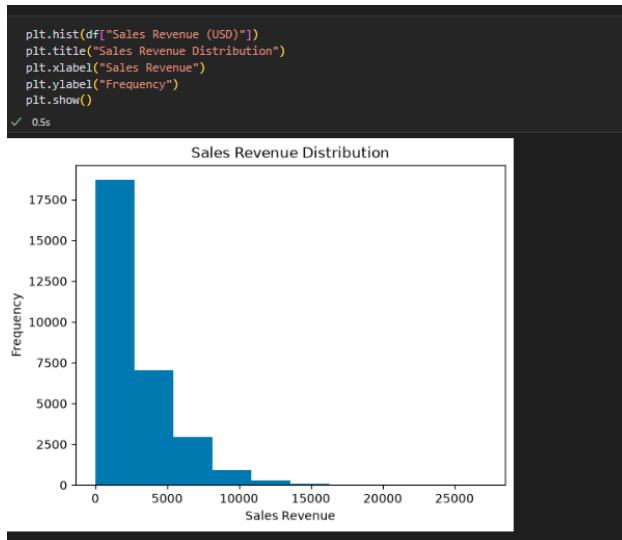
Matrix

```
df.corr(numeric_only=True)
```

[19] ✓ 0.3s

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Holiday Effect
Product ID	1.000000	-0.008570	-0.181934	-0.006891	-0.002496	-0.000015
Units Sold	-0.008570	1.000000	0.615767	-0.001620	-0.004750	0.159324
Sales Revenue (USD)	-0.181934	0.615767	1.000000	-0.065791	-0.002732	0.090681
Discount Percentage	-0.006891	-0.001620	-0.065791	1.000000	-0.000322	0.005848
Marketing Spend (USD)	-0.002496	-0.004750	-0.002732	-0.000322	1.000000	-0.000455
Holiday Effect	-0.000015	0.159324	0.090681	0.005848	-0.000455	1.000000

Histogram



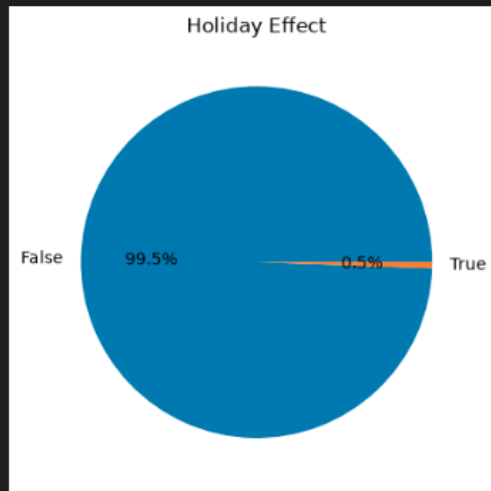
Bar Chart



Pie chart

```
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")  
plt.title("Holiday Effect")  
plt.ylabel("")  
plt.show()
```

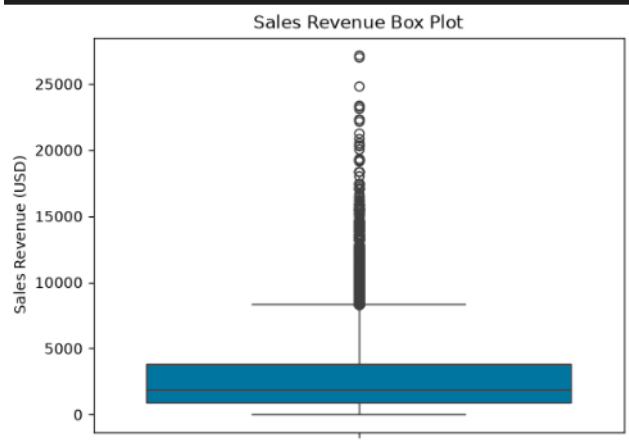
✓ 0.1s



Box plot

```
sns.boxplot(y=df["Sales Revenue (USD)"])  
plt.title("Sales Revenue Box Plot")  
plt.show()
```

✓ 0.1s



Correlation heatmap

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Blues")  
plt.title("Correlation Heatmap")  
plt.show()
```

✓ 0.3s

